🩸 Blood Sugar & Diabetes: Guided Notes · ANSWER KEY

Biology A · Unit 6 · Level 3 (closer to the class work) · goes with Unit 3 Diabetes: A1 CDC data, A3/A6 Homeostasis, ClEvR quizzes, A7 Risks, A8 Solutions, A10 Review, Diabetes Test

Name: _____

Use the class slides and your reading. Key words:

homeostasis | *blood glucose* | *insulin* | *glucagon* | *pancreas* | *feedback loop* | *type 1* | *type 2* | *risk factor*

A. Homeostasis (A3/A6)

1. Keeping inside conditions balanced is **homeostasis** .

2. Fill in the table.

Blood sugar is…	Hormone released	What happens
HIGH (after a meal)	**insulin**	**glucose goes into cells; blood sugar goes down**
LOW (skipped a meal)	**glucagon**	**stored sugar is released; blood sugar goes up**

B. Diabetes (ClEvR, A1)

3. Most cases (90-95%) are type **2** .

4. **Reading the ClEvR #2 graph**

> Example: 210 − 130 = 80 mg/dL higher

a) Diabetic: 240. Non-diabetic: 135. Difference = **105 mg/dL**

b) Which number is above the 200 line? **240 (diabetic)**

5. Why do sugary drinks spike blood sugar more than beans?
Sugary drinks have sugar with no fiber, so it enters the blood fast; beans have fiber and raise it slowly.

C. Risks and solutions (A7, A8)

6. Fill in the table.

Risk factor	Genetic or environmental?	Can it change?
family history	**genetic**	**no**
daily activity	**environmental**	**yes**
food in local stores	**environmental**	**yes, with community help**

7. Describe one community solution and how it lowers risk.
Example: water fountains at school so students drink water instead of soda, lowering sugar spikes.

🔑 *Key written by our team from Ms. Dunham's materials, please check.*